

Shital Sable

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SKILLS

Software Development: Python, ROS, C++, git, Linux, CI/CD, AI-Assisted Development (Cursor)
Perception & Sensors: LiDAR, Camera, Sensor Calibration, Sensor Fusion, Point Clouds, SLAM, Depth Estimation, SfM, Gaussian Splatting
Libraries and Models: PyTorch, OpenCV, Scikit-learn, Tensorflow, Pandas, NumPy, Matplotlib, SciPy, YOLO, SAM2, Object detection and Classification, Nerfstudio
Certifications: Deep Learning (deeplearning.ai), Kalman Filter(Udemy)

EXPERIENCE

Robotics Engineer at ROCK Robotic

April 2024 - Mar 2025

Skills: LiDAR, IMU, Linux, Git, Docker, CUDA, Python, C++, OpenCV, pix4d, Jira, Notion

California

- Designed and implemented improvements to LiDAR calibration pipelines; conducted QA on 300+ datasets, generating detailed reports.
- Developed and tested a Gaussian splat pipeline for 3D reconstruction, refining over 30 datasets to improve the algorithm. Integrated YOLO-based segmentation into perception pipeline for object masking and packaging.
- Contributed to SLAM and state estimation systems in large, peer-reviewed C++/Python codebases.
- Evaluated algorithms including SfM, depth estimation, and novel view synthesis.

Research Intern at LifePlus (Startup)

Nov 2023 - Feb 2024

Skills: Sensor Fusion, Signal analysis, Python, Scipy, Numpy, Linear Algebra, PPG, Accelerometer

San Jose, CA

- Led sensor fusion and signal processing for PPG and Accelerometer data.
- Reduced error margins by 75% using Singular value Decomposition and FFT for motion artifact removal.
- Reviewed 30+ research papers to benchmark motion artifact mitigation in wearable sensors.

Software Engineer, Robotics at Oregon State University

Jul 2023 - Jul 2024

Skills: Tensorflow, Point Clouds, Keras, Linux, Git

Corvallis, Oregon

- Analyzed 3 methods to implement classification based on 'Function Basis Modelling' for 3D point cloud of regular objects
- Implemented Multi-Label classification on top of PointNet to achieve the point cloud data classification
- Designed a custom encoding function to handle the classes and weighted binary cross entropy function to handle the data bias in the model

Robotics Engineer - Graduate Assistant at Oregon State University

Jan 2022 - June 2023

Skills: IMU, Motion Capture, State Estimation, Kalman Filter

Corvallis, Oregon

- Investigated IMU-based state estimation for deformable sensor arrays; surveyed 5 advanced estimation methods.
- Contributed to underwater docking platform development, including system design and actuator programming.
- Assessed 130+ research articles for Underwater Motion Capture and State Estimation system

Software Developer - Graduate Assistant at Oregon State University

Oct 2019 - Mar 2022

Skills: Linear Algebra, Convolution, Matlab, Machine Learning

Corvallis, Oregon

- Developed a physics simulator for electric space propulsion; optimized performance using convolution (300% speedup).
- Created a neural network model trained on 1000+ simulated data points, achieving 99% accuracy.

Design Engineer at Arcatron Mobility(Startup)

Jul 2017 - Nov 2018

Skills: Inventor(3D CAD), AutoCAD

Pune, India

- Designed a shower-commode wheelchair frame to improve stability and comfort while being suitable for nonstandard conditions.
- Performed quality inspection of prototype wheelchairs and sub-assemblies before deployment Generating documentation for best practices

COMPETITIONS

Chassis Lead at SAE BAJA (Team Read Baron)

Mar 2015-Jul 2017

Skills: Catia, ANSYS, Hypermesh

India

- Designed, analyze and fabricate components of a prototype All-Terrain Vehicle including steering, brake, gearbox and suspension assembly in a team of 40 undergraduate students
- Manufactured a Tubular Space-frame and the 5 fixture assemblies for optimize Chassis manufacturing and assembly
- Reduced the manufacturing error margin by 70% by improving the fixture design and leveraging surface modelling methods

SELECTED PROJECTS

Two-Step Learning-Based Path-Following Control for a Truck-Trailer at Learning Based Control

OSU

Skills: Python, git, Linux

- Combined 2 learning methods: supervised and unsupervised learning to generate a neural network for improving path planning of truck trailer
- Trained neural networks to minimize trajectory deviation under kinematic constraints

Autonomous Mobile Robot at

Skills: Python, ROS, slam toolbox, Cartographer, Nav2 Hardware: Raspberry Pi, Stereo Camera, Lidar

- Designed the Robot in simulation environment
- Subsequent parts will include designing steering and control, teleop, mapping and localization with SLAM, autonomous navigation and object tracking

Hat following Tello Drone at Intro to Robotics

OSU

Skills: Python, opencv, git. Hardware: DJI Tello Drone

- Tracked a colored hat and estimated its coordinates using OpenCV in Python
- Estimated the pose of an object using ArUCo Markers
- Implemented proportional control maintain the pose and distance of drone with a hat

EDUCATION

Masters in Robotics Engineering, Oregon State University

Jan 2022 - Jun 2023

Bachelors in Mechanical Engineering, University of Pune

Aug 2013 - Apr 2017

PUBLICATIONS

- [1] S. D. Sable and B. Camburn, "Topology generation for ion-thruster grids," *AIAA Propulsion and Energy 2020 Forum*, Jul. 2021. [Online]. Available: <https://doi.org/10.2514/6.2021-3419>
- [2] Z. A. Connor, S. Sable, M. I. Campbell, and R. B. Stone, "Automated functional basis modeling using multi-label machine learning on 3d models," *ASME 2024 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Aug. 2024. [Online]. Available: <https://doi.org/10.1115/detc2024-138095>
- [3] P. G. G. Momin, S. Sable, A. Tayade, R. Thube, and T. Gedam, ""design, analysis and fabrication of a rollcage for an all-terrain vehicle"," vol. 3, p. 201–207, Nov. 2021. [Online]. Available: <https://www.ijtimes.com/index.php/ijtimes/article/view/1204>